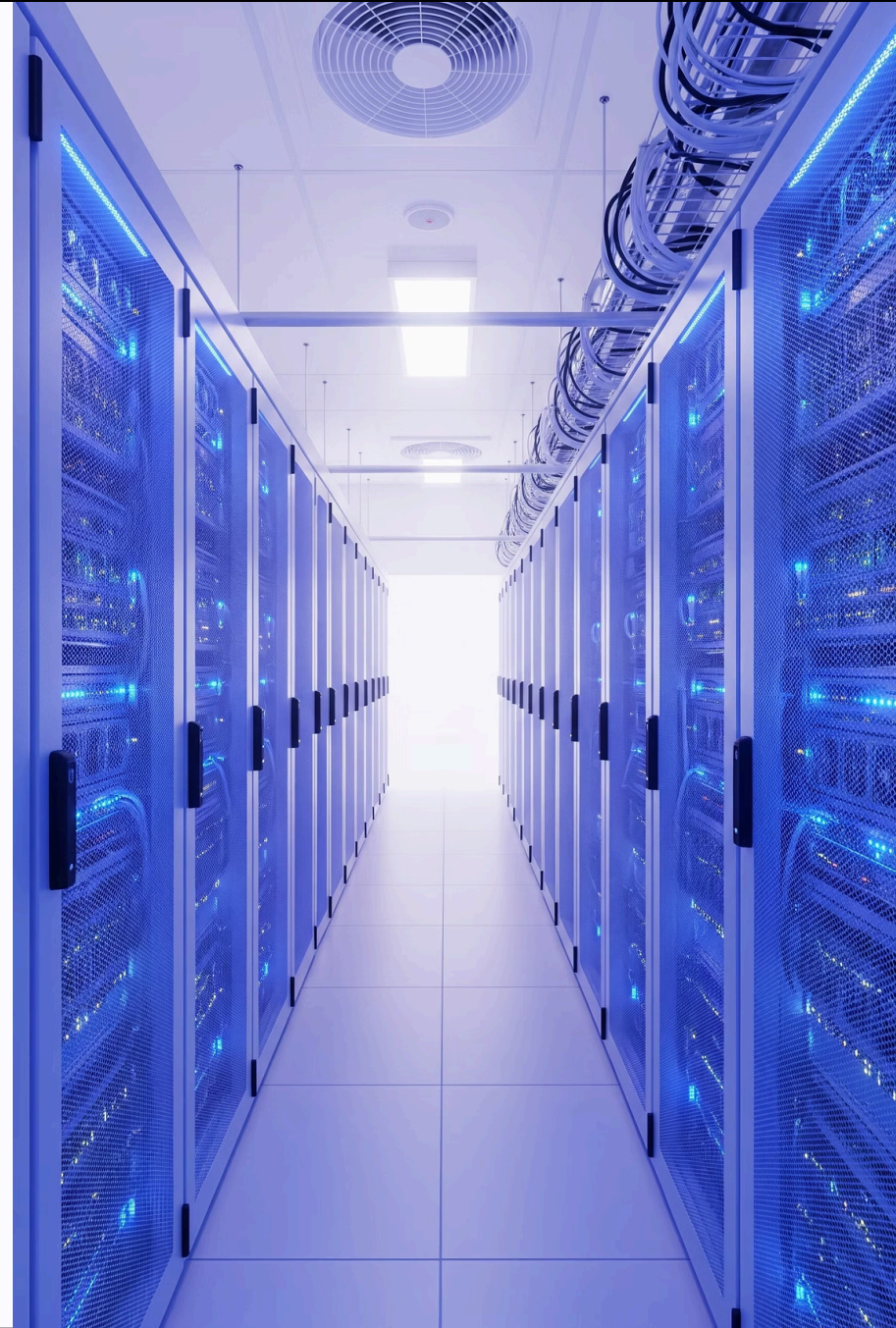


Review of Cloud Infrastructure for AI

Cloud infrastructure has become the backbone of AI workloads, enabling **scalability, elasticity, and global reach** that would be impossible with on-premises solutions alone.

This presentation explores the key components of cloud infrastructure for AI: compute, storage, networking, and orchestration.



Why Cloud for AI?

Training large language models requires thousands of GPUs running for weeks—a task only feasible in cloud environments. Similarly, inference operations must handle billions of daily API requests with varying load patterns.

Pay-as-you-go Scaling

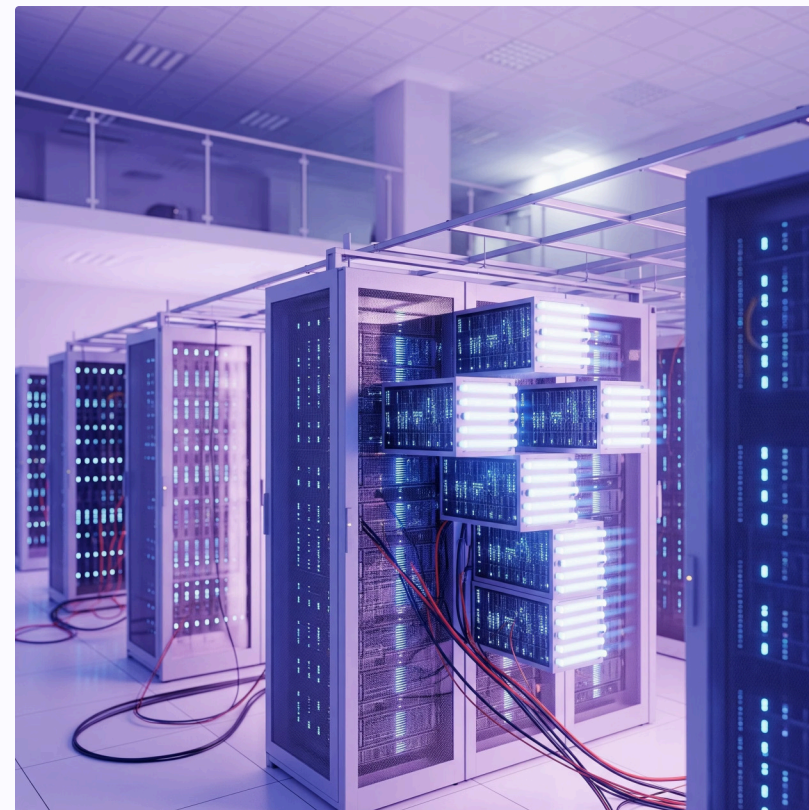
No need to invest in expensive hardware that sits idle between training runs

Global Data Center Coverage

Deploy AI services closer to users for reduced latency and improved performance

Managed AI Services

Focus on model development rather than infrastructure management



Core Cloud Providers

The major cloud providers are competing aggressively in the AI space, each with unique strengths and specialized offerings.

AWS

- SageMaker comprehensive ML platform
- Trainium/Inferentia custom AI chips
- Vast infrastructure ecosystem

Azure

- Exclusive OpenAI partnership
- Enterprise AI focus
- Tight Microsoft ecosystem integration

Google Cloud

- TPU pods for specialized ML workloads
- Vertex AI unified platform
- BigQuery ML for data-centric AI

All providers are increasingly competing on **AI-native services** and custom silicon designed specifically for machine learning workloads.

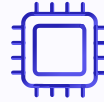
Compute Services

Cloud providers offer multiple compute options optimized for different AI workloads, from training to inference to batch processing.



Virtual Machines

EC2, GCE, Azure VMs for general workloads with flexible configurations



GPU/TPU Instances

Specialized accelerators for deep learning with massive parallel processing capabilities



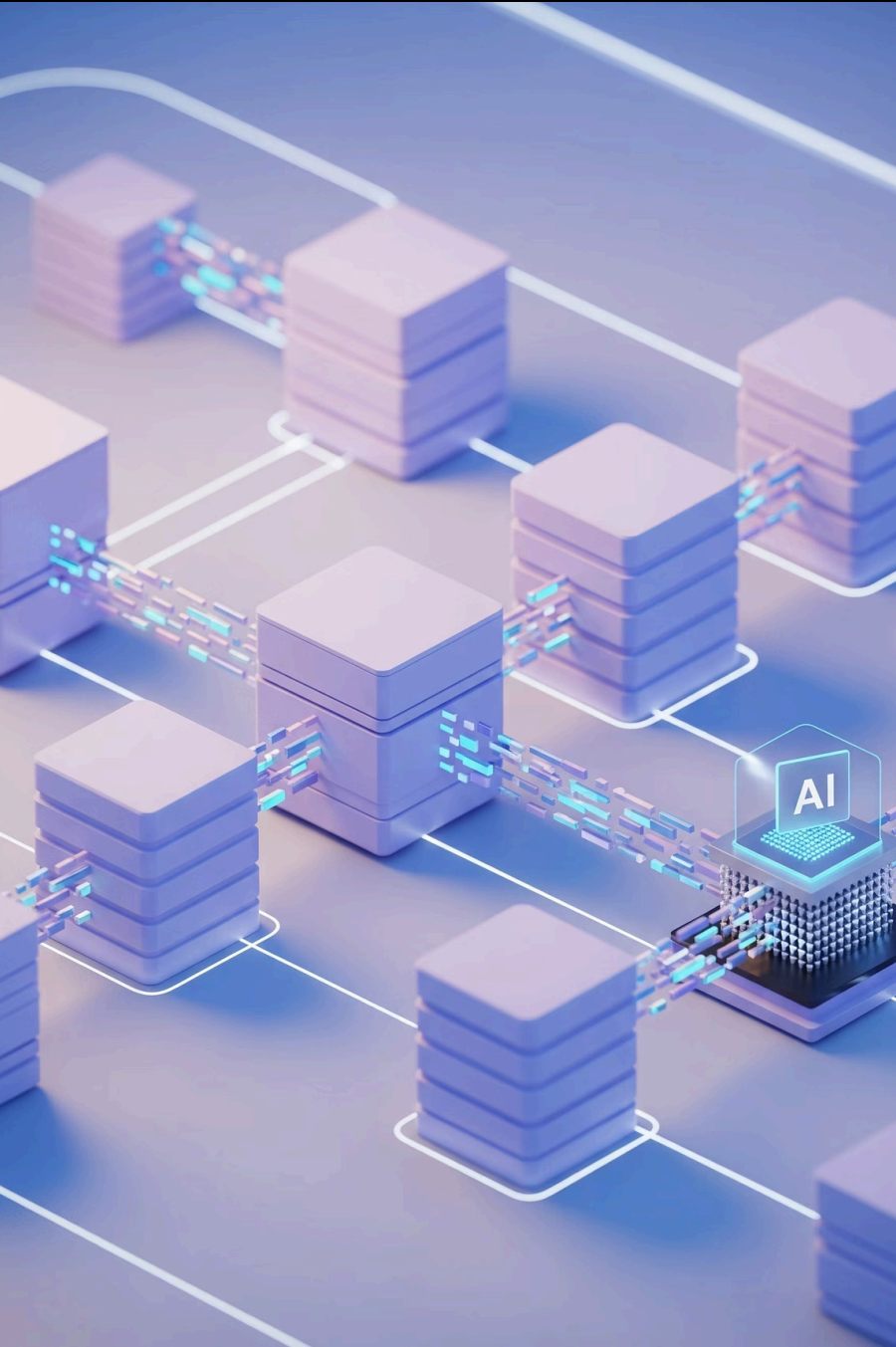
Serverless Compute

Lambda, Cloud Functions for event-driven ML tasks without server management



Containers

ECS, GKE, AKS for deploying AI microservices with consistent environments



Storage Services

Object Storage

S3, GCS, Azure Blob Storage provide nearly unlimited capacity for datasets, model artifacts, and training data with high durability

Block Storage

EBS, Persistent Disks attach directly to training nodes for low-latency file operations during model training

Parallel File Systems

FSx Lustre, GCS Filestore enable high-throughput distributed training across multiple GPU nodes



Data locality is critical for AI efficiency. Placing storage close to compute resources minimizes data transfer times and maximizes throughput during training.

Networking in Cloud AI

High-performance networking is the foundation of distributed AI training and global inference deployments.

Virtual Private Clouds (VPCs)

Isolate AI workloads securely with private network segments and security groups

Specialized Interconnects

InfiniBand, NVLink for GPU clusters enable near-zero latency between compute nodes

Content Delivery Networks

Globally distributed edge locations for low-latency AI inference APIs



Latency-sensitive AI applications require strategic placement near users to minimize response times and improve user experience.

Orchestration & MLOps

Kubernetes

EKS, GKE, AKS provide container orchestration for model deployment and scaling

Monitoring

Track model performance, drift, and system health in production



ML Pipelines

Kubeflow, MLflow, Vertex AI Pipelines automate training workflows

CI/CD for ML

Automated testing, retraining, and deployment of models in production

Cloud-native MLOps frameworks standardize the AI lifecycle, enabling teams to consistently deliver models from research to production.

Cost & Scaling Strategies

AI workloads can quickly become expensive without proper cost management strategies.

Spot/Preemptible Instances

Reduce GPU costs by 70-90% for fault-tolerant training jobs using excess capacity

Auto-scaling Clusters

Dynamically adjust inference capacity based on actual usage patterns and traffic

Hybrid Cloud Approaches

Balance sensitive data requirements with elasticity needs across on-prem and cloud

 **FinOps for AI** applies cloud cost governance specifically to machine learning workloads, tracking spending by model, team, and business outcome.

Security & Compliance

Identity & Access Management

Fine-grained control over who can access AI resources, training data, and model artifacts

Encryption

Protect sensitive data and models both at rest in storage and in transit across networks

Regional Compliance

Address GDPR, HIPAA, and emerging AI regulations through proper geographic data placement

Cloud Security Posture

Continuously monitor and enforce security policies specific to AI workloads

Security considerations become even more critical as AI systems access sensitive data and make high-impact decisions.



Edge & Hybrid Cloud



Hybrid Model

On-premises for sensitive data processing with cloud for large-scale training



Edge AI

Inference on NPUs/Jetson devices with models trained in cloud



5G Integration

Low-latency connectivity enabling real-time robotics and autonomous vehicles

The convergence of cloud and edge is an accelerating trend, with models trained centrally but deployed globally at the network edge.

Monitoring & Observability

Comprehensive monitoring is essential for maintaining reliable AI systems and detecting issues before they impact users.

Infrastructure Monitoring

CloudWatch, Stackdriver, Azure Monitor track system health, utilization, and bottlenecks

AI-Specific Metrics

DCGM for GPU health, custom Prometheus/Grafana dashboards for model performance

Drift Monitoring

Detect when production data patterns diverge from training data, potentially requiring retraining

Audit Logging

Comprehensive compliance logging and audit trails for model decisions and actions

Case Study: OpenAI on Azure

The Infrastructure Behind GPT

- Azure built custom **AI supercomputers** with 10,000+ GPUs interconnected for distributed training
- Global deployment enables GPT API responses in under 200ms latency for users worldwide
- Tight integration with enterprise systems through Microsoft Copilot and Dynamics

"The scale of cloud infrastructure required for modern AI models was unimaginable just five years ago. What OpenAI has built on Azure demonstrates how cloud infrastructure has become the essential product enabler for AI companies."



Cloud AI Infrastructure Pipeline

Cloud AI infrastructure = data, compute, orchestration, serving, compliance.

This complete pipeline demonstrates how modern AI systems leverage cloud infrastructure at every stage from data preparation through deployment and monitoring.

Key Lessons

1

Cloud-First AI

Cloud has become the default home for AI scaling, enabling capabilities impossible with on-premises infrastructure

2

Locality Matters

Compute and storage locality is critical for performance, with specialized interconnects reducing data transfer bottlenecks

3

Multi-Cloud Reality

Multi-cloud and hybrid strategies are emerging as organizations balance performance, compliance, and vendor lock-in concerns

4

Built-in Compliance

MLOps and compliance frameworks are increasingly built directly into cloud-native AI platforms

Discussion

If you were CTO of a healthcare AI startup, would you go all-in on cloud GPUs or adopt a hybrid cloud with local data residency — and why?

Consider factors like:

- HIPAA compliance and patient data privacy
- Training performance requirements
- Cost implications of different architectures
- Inference latency needs for clinical applications

